



Excess molar enthalpies and excess molar heat capacities of (chloroform + tetrahydrofuran + cyclohexane) at the temperature 298.15 K

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Excess molar enthalpies and excess molar heat capacities at $T = 298.15$ K were determined for $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O or } c\text{-C}_6\text{H}_{12}\}$ and for $\{x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_4\text{O} + (1-x_1-x_2)c\text{-C}_6\text{H}_{12}\}$. Both experimental properties were represented by a Redlich–Kister type equation. The large negative values of the excess enthalpies and the positive values of the excess heat capacities observed in the ternary composition domain are mainly determined by the specific interaction between chloroform and tetrahydrofuran. The properties of the ternary system were poorly reproduced by combining the information of the parent binary mixtures using empirical rules. They were, however, better described by semi-empirical equations such as the Wang, UNIQUAC, and UNIFAC methods. © 1997 Academic Press Limited

KEYWORDS: excess enthalpies; excess heat capacities; ternary mixture; polar

1. Introduction

The study of the excess thermodynamic properties of some multicomponent mixtures made up of components with different polar character are in progress. So far we have determined the excess enthalpies and excess heat capacities of all the binary and ternary combinations of the compounds: ethanol, dimethylformamide, tetrahydrofuran, and cyclohexane.^(1–4) Measurements of the excess Gibbs energies and volumes for the same mixtures have either already been carried out,⁽⁵⁾ or are in progress. Experimental measurements of the excess enthalpies and excess heat capacities for some quaternary mixtures of these compounds have also been examined.⁽⁶⁾

In order to take into consideration other compounds which display different types of interactions, we chose to add to the above list chloroform, which is known to show hydrogen bonding interactions with the cyclic ether.^(7,7a) We report the results of measurements of H_m^E and $C_{p,m}^E$ at $T = 298.15$ K for (chloroform + tetrahydrofuran + cyclohexane) and for two of the parent binary mixtures.

The experimental results were represented by a polynomial equation. The values of the thermodynamic properties of the ternary mixtures were tentatively reproduced by combining the properties of the binary pairs using common empirical rules. The enthalpy values were also compared with values predicted by the UNIQUAC, UNIFAC, and Wang predictive equations.

2. Experimental

Excess molar enthalpies and excess molar heat capacities were measured by means of a flow calorimeter following the experimental procedure as previously described.⁽¹⁾ Further details on the calibration of the apparatus and on the reference chosen for the heat capacity measurements (normally *n*-heptane, used as carrier) have also been previously reported.⁽²⁾ A supplementary cooling system (thermostatted cool water from an external cryothermostat driven by an h.p.l.c. pump into an Inconel tubing surrounding the mixing cell) was introduced into the calorimeter for enthalpy measurements to compensate for the large exothermic effects associated with the mixing of chloroform with tetrahydrofuran.

The measurements of the data concerning the ternary mixtures were obtained as follows: in the experiments measuring enthalpy of mixing, cyclohexane was mixed with $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$, whereas the heat capacity data were obtained for a series of pseudobinary mixtures in which chloroform was added to $\{xc\text{-(CH}_2)_4\text{O} + (1-x)c\text{-C}_6\text{H}_{12}\}$.

The densities of the mixtures at $T = 298.15\text{ K}$, used for the heat capacity determinations, were calculated from experimental density values of the pure compounds measured with a DMA-60 Anton-Paar vibrating-tube densimeter equipped with a DMA602 cell, and from excess volumes determined previously by Conti *et al.*⁽⁸⁾

The reagents (mass fraction: 0.995) were used without further purification. Their purity was checked by gas-chromatography. *n*-Heptane and anhydrous ethanol (sometimes used as a reference liquid for heat capacity measurements) were from Carlo Erba; all other reagents were from Fluka. Tetrahydrofuran was maintained over molecular sieves and chloroform was stabilized with 10 ppm amylene. The water content (by the Karl-Fischer method) was 0.01 per cent for $c\text{-(CH}_2)_4\text{O}$ and 0.026 per cent for CHCl_3 . The densities and molar heat capacities of the used compounds at $T = 298.15\text{ K}$ were: for CHCl_3 , $1.47910\text{ g}\cdot\text{cm}^{-3}$ and $113.93\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$; for $c\text{-(CH}_2)_4\text{O}$, $0.88219\text{ g}\cdot\text{cm}^{-3}$ and $122.59\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$; for $c\text{-C}_6\text{H}_{12}$, $0.77389\text{ g}\cdot\text{cm}^{-3}$ and $156.01\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$; for $\text{CH}_3(\text{CH}_2)_5\text{CH}_3$, $0.67951\text{ g}\cdot\text{cm}^{-3}$ and $224.77\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$; and for $\text{CH}_3\text{CH}_2\text{OH}$, $0.78540\text{ g}\cdot\text{cm}^{-3}$ and $111.91\text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$.

3. Results

Experimental values of the molar excess enthalpies H_m^E and molar excess heat capacities $C_{p,m}^E$ for the binary mixtures studied in this work are reported in

TABLE 1. Excess molar enthalpies H_m^E for binary mixtures at $T = 298.15$ K for different mole fractions x

x	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$	x	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$
$\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$							
0.1012	-810	0.3027	-2177	0.5032	-2767	0.9012	-908
0.1012	-813	0.3027	-2157	0.6031	-2633	0.9012	-910
0.1012	-810	0.4031	-2583	0.7027	-2289		
0.2021	-1579	0.5032	-2742	0.7027	-2286		
0.2021	-1553	0.5032	-2707	0.8020	-1681		
$\{x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}\}$							
0.0563	150.8	0.2666	519.0	0.4732	644.3	0.7587	478.3
0.0563	147.4	0.2666	520.2	0.5740	618.7	0.8435	351.5
0.1643	375.2	0.3661	609.6	0.5740	618.1	0.9238	199.7
0.2520	500.1	0.4732	630.9	0.6690	567.7		

tables 1 and 2, respectively. Both properties were represented by the equation:

$$Y_{ij,m}^E = x_i x_j \cdot \sum_{k=1}^5 A_k (x_i - x_j)^{k-1}, \quad (1)$$

where the subscripts i and j ($i < j$) refer to the pure components: 1 represents CHCl_3 ; 2, $c\text{-(CH}_2)_4\text{O}$; and 3, $c\text{-C}_6\text{H}_{12}$; and where $Y_{ij,m}^E$ is either $H_m^E/(\text{J}\cdot\text{mol}^{-1})$ or $C_{p,m}^E/(\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})$. Table 3 contains the values of the parameters A_k and the standard deviation of the fit. A non-linear least-squares fit was used for this purpose. In order to give a more balanced weight to the small H_m^E values of the dilute mixtures, the function $H_m^E/x_i x_j$ was selected in the minimization procedure for the enthalpy data.

The H_m^E and $C_{p,m}^E$ data for the binary systems involving chloroform are shown in figures 1 and 2, together with data from the literature. The enthalpy data relating

TABLE 2. Excess molar heat capacity $C_{p,m}^E$ for binary mixtures at $T = 298.15$ K for different mole fractions x

x	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$	x	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$	x	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$	x	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$
$\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$							
0.1002	1.60	0.3990	7.41	0.5999	8.64	0.8024	4.75
0.2001	3.55	0.4022	7.30	0.6139	8.37	0.8997	2.17
0.2067	3.56	0.4781	8.64	0.7003	7.10	0.9025	2.42
0.3002	5.62	0.4997	8.47	0.7038	7.10	0.9053	2.32
0.3082	5.50	0.5099	8.68	0.7799	5.49	0.9497	1.10
$\{x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}\}$							
0.1000	-0.67	0.3889	-1.43	0.5682	-1.54	0.7497	-1.11
0.2098	-1.10	0.4662	-1.52	0.5714	-1.58	0.8384	-0.94
0.3098	-1.41	0.5006	-1.48	0.7015	-1.32	0.9107	-0.53

TABLE 3. Parameters A_k for representing the excess molar heat capacities $C_{p,m}^E$ and the excess molar enthalpies of binary mixtures; σ is the standard deviation

Property	Mixture	A_1	A_2	A_3	A_4	A_5	$\sigma/(\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})$
$C_{p,m}^E/(\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})$	$\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$	33.968	11.181	-20.763	-8.4951		0.17
	$\{x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}\}$	-6.1283		-1.3147			0.054
	$\{xc\text{-(CH}_2)_4\text{O} + (1-x)c\text{-C}_6\text{H}_{12}\}^a$	-4.0320	2.6898	-2.3931			0.04
$H_m^E/(\text{J}\cdot\text{mol}^{-1})$	$\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$	-10949	-810.26	2177.2			14
	$\{x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}\}$	2545.9	-125.09	361.22	197.83		4
	$\{xc\text{-(CH}_2)_4\text{O} + (1-x)c\text{-C}_6\text{H}_{12}\}^a$	2974.4	-161.33			226.11	3.2

^a Reference 1.

to the mixtures of tetrahydrofuran (figure 1) are in good agreement with data from other authors,⁽⁹⁻¹²⁾ particularly with those by Inglese.⁽⁹⁾ Not included in figure 1, but almost coincident with ours, are the data by Van Ness *et al.*⁽¹⁰⁾ when extrapolated to $T = 298.15$ K from the experimental temperatures of 303.15 K and 318.15 K. The only available literature $C_{p,m}^E$ data⁽¹³⁾ are lower than ours, displaying a value of $7.51 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ against our value of $8.49 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ for the composition of $x = 0.5$. We have no explanation for this discrepancy; a few more measurements using a chloroform sample preserved on molecular sieves yielded the same results.

The excess molar enthalpy data of the mixtures of chloroform with cyclohexane (figure 2) show a good agreement with those of Nagata⁽¹⁴⁾ and of Murakami,⁽¹²⁾ but deviate clearly from those of Findlay.⁽¹⁵⁾ No $C_{p,m}^E$ data for this binary are available for comparison.

The system $\{xc\text{-(CH}_2)_4\text{O} + (1-x)c\text{-C}_6\text{H}_{12}\}$ was studied previously by us,⁽¹⁾ and our previous measurements were confirmed by the new experimental data. The perfect agreement with earlier data allows us to use the analytical description (table 3) of H_m^E and $C_{p,m}^E$ of the previously reported mixtures.⁽¹⁾

The experimental values of H_m^E and $C_{p,m}^E$ for the ternary mixture are reported in tables 4 and 5, respectively. The values were fitted with the equation:

$$Y_{123,m}^E = x_1 x_2 x_3 \cdot \sum_{m=0}^3 \cdot \sum_{n=0}^m B_{mn} x_1^m x_2^n + \sum_{i=1}^2 \cdot \sum_{j=i+1}^3 Y_{ij,m}^E, \quad (2)$$

where $Y_{ij,m}^E$ are the corresponding binary excess properties obtained from equation (1) using the mole fractions of the ternary mixture. The best values of the parameters B_{mn} as well as the standard deviations σ of the fits are reported in table 6. We would like to point out that the standard deviations for both binary and ternary mixtures are, in general, within the experimental uncertainty. The larger σ value of the H_m^E for the ternary system cannot be significantly lowered by increasing the number of B_{mn} parameters because of the larger uncertainty (about 1 per cent) of the corresponding experimental H_m^E data. The trend of the excess functions is shown in figure 3.

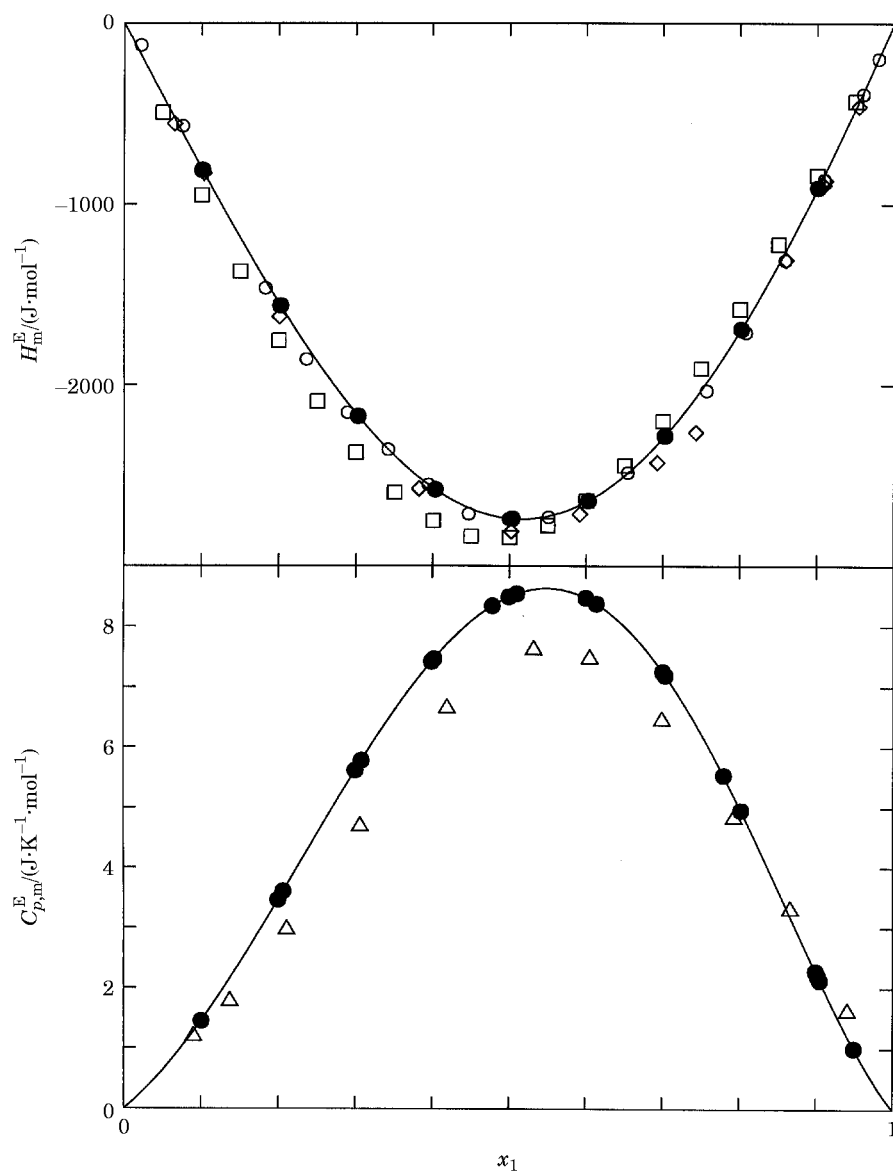


FIGURE 1. Excess molar enthalpies H_m^E and excess molar heat capacities $C_{p,m}^E$ for $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$ at $T = 298.15\text{ K}$: —, best-fitting curve from equation (1); ●, this work; □, Becker *et al.*;⁽¹¹⁾ ◇, Murakami *et al.*;⁽¹²⁾ ○, Inglese;⁽⁹⁾ △, Inglese *et al.*⁽¹³⁾

4. Discussion

Figure 3 shows that the excess molar heat capacities of (chloroform + tetrahydrofuran + cyclohexane) are almost always positive while the excess enthalpies are mostly negative and change sign only in the hydrocarbon-rich region.

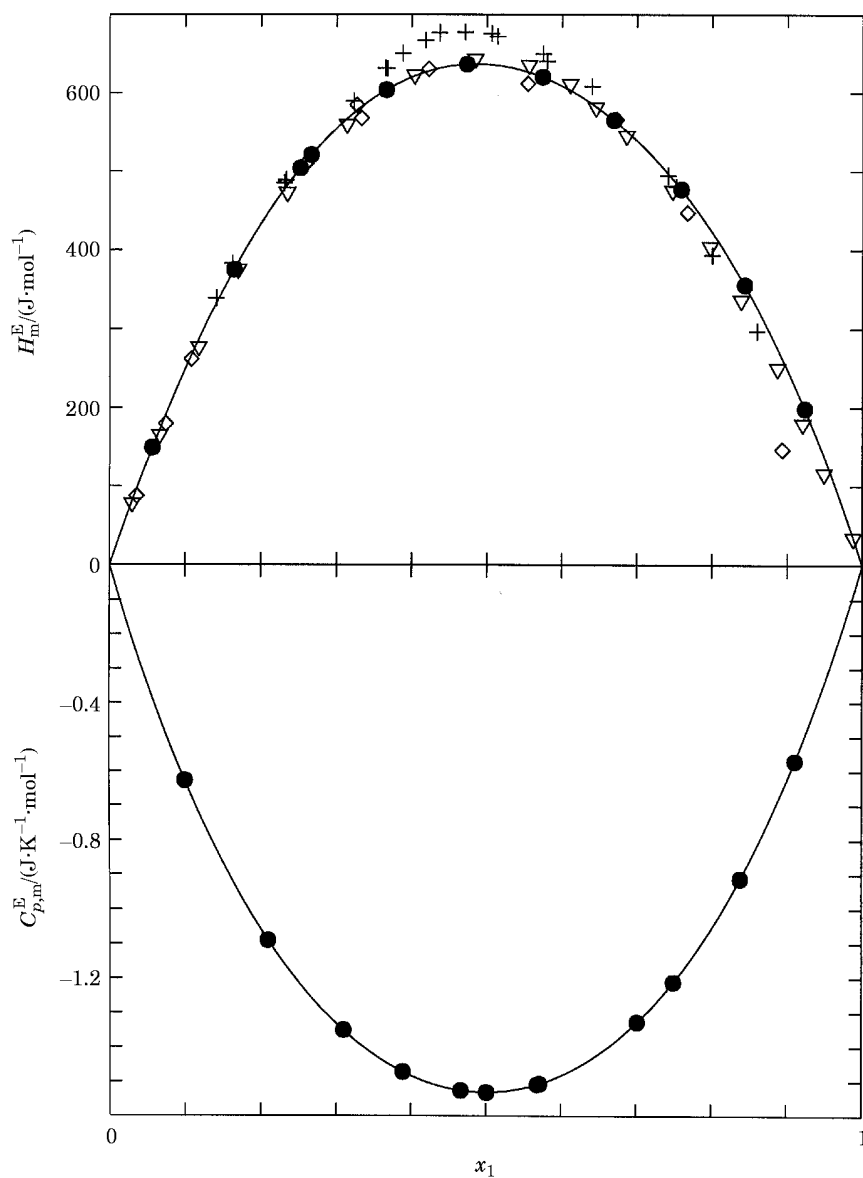


FIGURE 2. Excess molar enthalpies H_m^E and excess molar heat capacities $C_{p,m}^E$ for $\{x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}\}$ at $T = 298.15\text{ K}$: —, best-fitting curve from equation 1; ●, this work; ▽, Nagata *et al.*,⁽¹⁴⁾ ◇, Murakami *et al.*,⁽¹²⁾ +, Findlay *et al.*⁽¹⁵⁾

The large and negative enthalpy values clearly reflect the formation of hydrogen bonds between the CH of chloroform and the ether oxygen. The positive heat capacity values are on the other hand consistent with the large relaxation effect associated with the breaking of these bonds as the temperature is raised during the experimental determination of this property. The sign of H_m^E and $C_{p,m}^E$ is consistent

TABLE 4. Excess molar enthalpies H_m^E of $\{x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_6\text{O} + (1-x_1-x_2)c\text{-C}_6\text{H}_{12}\}$ at $T = 298.15$ K

x_1	x_2	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$	x_1	x_2	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$	x_1	x_2	$\frac{H_m^E}{\text{J}\cdot\text{mol}^{-1}}$
0.0925	0.8306	-524	0.3372	0.5058	-1744	0.3311	0.1421	-243
0.0844	0.7577	-280	0.3032	0.4548	-1358	0.4681	0.2009	-835
0.0758	0.6810	-57.2	0.2673	0.4009	-987	0.2561	0.1099	-11.4
0.0668	0.5999	126.1	0.2292	0.3439	-654	0.5308	0.2278	-1170
0.0573	0.5142	280	0.1889	0.2834	-326	0.1763	0.0757	137.2
0.0472	0.4234	388	0.1461	0.2191	-66.9	0.5902	0.2533	-1531
0.0364	0.3272	438	0.1005	0.1508	117.9	0.0911	0.0391	177.2
0.0250	0.2250	432	0.2292	0.3439	-650	0.6464	0.2774	-1908
0.0129	0.1161	303	0.3032	0.4548	-1366	0.4016	0.1724	-509
0.0472	0.4234	380	0.4619	0.4618	-2312	0.4591	0.1149	-222
0.0668	0.5999	130.4	0.4217	0.4215	-1902	0.3785	0.0948	-18.1
0.1847	0.7385	-1213	0.3792	0.3791	-1490	0.5351	0.1339	-474
0.1686	0.6738	-900	0.3343	0.3342	-1092	0.2928	0.0733	133.8
0.1515	0.6057	-615	0.2868	0.2867	-733	0.6068	0.1519	-744
0.1335	0.5337	-353	0.2364	0.2363	-411	0.2016	0.0504	229
0.1144	0.4576	-113.9	0.1828	0.1828	-130.1	0.6746	0.1689	-1041
0.0943	0.3769	82.5	0.1258	0.1258	83.7	0.7371	0.1867	-1382
0.0729	0.2913	223	0.0650	0.0650	142.3	0.1039	0.0263	204
0.0501	0.2004	292	0.3792	0.3791	-1497	0.4580	0.1160	-248
0.1335	0.5337	-363	0.2364	0.2363	-414	0.3776	0.0956	-33.2
0.2774	0.6460	-1772	0.5545	0.3692	-2227	0.5165	0.0575	168.4
0.2532	0.5896	-1410	0.5063	0.3371	-1821	0.4258	0.0474	281
0.2276	0.5301	-1065	0.4553	0.3032	-1427	0.6020	0.0670	-3.0
0.2006	0.4672	-744	0.4015	0.2673	-1043	0.3294	0.0367	343
0.1720	0.4006	-447	0.3445	0.2293	-700	0.6827	0.0760	-186.0
0.1417	0.3301	-178.5	0.2839	0.1891	-368	0.2268	0.0252	344
0.1096	0.2552	52.4	0.2196	0.1463	-104.2	0.7590	0.0845	-411
0.0389	0.0907	215	0.1512	0.1006	89.9	0.1172	0.0130	246
0.1720	0.4006	-438	0.0781	0.0520	160.4	0.8313	0.0926	-650
0.3694	0.5541	-2149	0.4016	0.1724	-517			

with the qualitative scheme proposed by Patterson,⁽¹⁶⁾ according to which the creation of a larger order in the mixtures with respect to the pure components would involve negative values for enthalpy, Gibbs energy and entropy, but positive values for heat capacity.

The shapes of the H_m^E and $C_{p,m}^E$ surfaces over the ternary composition space is dominated by chloroform–tetrahydrofuran interactions. Both surfaces appear symmetrical with respect to the diagonal of the equimolar mixtures of these compounds, exhibiting a smooth decrease of the large magnitude effects typical of this binary as the mixtures become richer in cyclohexane. In other words, addition of cyclohexane appears simply to dilute the C-H...O hydrogen bonds adding no new specific interaction.

We have examined the possibility of predicting the H_m^E and $C_{p,m}^E$ of the present ternary mixtures from information on the parent binary mixtures using both empirical and theoretical methods.

The empirical calculation of $Y_{123,m}^E$ through equation (2) is based on neglecting the first term on the right-hand side of equation (2) and using different procedures for

TABLE 5. Excess molar heat capacities $C_{p,m}^E$ of $\{x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_6\text{O} + (1 - x_1 - x_2)c\text{-C}_6\text{H}_{12}\}$ at $T = 298.15\text{ K}$

x_1	x_2	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$	x_1	x_2	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$	x_1	x_2	$\frac{C_{p,m}^E}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$
0.1494	0.0540	1.38	0.7432	0.0929	2.05	0.5238	0.3577	7.90
0.3720	0.0399	0.47	0.9011	0.0358	0.47	0.4204	0.4353	7.80
0.2668	0.0466	0.90	0.8191	0.0654	1.32	0.7984	0.1856	4.69
0.6662	0.0212	-0.42	0.6484	0.1309	3.26	0.7020	0.2742	6.73
0.5691	0.0274	-0.31	0.1306	0.4906	2.55	0.6070	0.3617	8.01
0.0730	0.0655	1.21	0.2350	0.4290	5.19	0.4083	0.5445	7.56
0.9122	0.0055	-0.32	0.3436	0.3692	6.89	0.3129	0.6323	5.89
0.1418	0.1103	2.74	0.4310	0.3214	7.16	0.2188	0.7189	3.75
0.2546	0.0959	2.68	0.5282	0.2677	6.65	0.1172	0.8124	1.56
0.3609	0.0821	2.05	0.6187	0.2143	5.33	0.5047	0.4381	8.35
0.4732	0.0677	1.24	0.8043	0.1084	2.54	0.0635	0.8477	0.62
0.5688	0.0554	0.69	0.0618	0.5276	0.72	0.0420	0.8682	0.21
0.6577	0.0440	0.46	0.7174	0.1527	3.69	0.5608	0.1084	2.80
0.0483	0.1309	1.12	0.2299	0.5784	4.64	0.4589	0.1335	3.65
0.0715	0.1222	1.67	0.1258	0.6566	2.06	0.3570	0.1587	4.30
0.7474	0.0321	0.12	0.0748	0.6890	0.78	0.2442	0.1865	4.59
0.1439	0.3096	3.53	0.0499	0.7057	0.36	0.1365	0.2131	3.38
0.2522	0.2704	5.50	0.3291	0.5037	6.50	0.2000	0.3601	4.70
0.3551	0.2332	5.87	0.9000	0.0751	1.77	0.2001	0.4388	4.45
0.5553	0.1608	4.15	0.8186	0.1362	3.37	0.5014	0.4476	8.45
0.5274	0.1709	4.37	0.7108	0.2172	5.30			
0.0873	0.3372	1.91	0.6330	0.2757	6.90			

the evaluation and weighting of the binary $Y_{ij,m}^E$ terms. We have tested a series of mixing rules which have been summarized by Lopez *et al.*⁽¹⁷⁾ and Pando *et al.*⁽¹⁸⁾ These authors also provide references to the literature. The empirical rules were judged according to the root mean square (r.m.s.) deviation and the mean deviation between the experimental values of the thermodynamic properties and the values calculated by each single method. We have also tested the effect of rotating the components, as asymmetric equations give more weight to binary contributions 1–2

TABLE 6. Ternary parameters B_{mm} of equation (2) with the standard deviation $\sigma/(\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1})$ for $C_{p,m}^E$, and $\sigma/(\text{J}\cdot\text{mol}^{-1})$ for H_m^E

	$\frac{B_{mm}\cdot 10^{-5}{}^a}{\text{J}\cdot\text{mol}^{-1}}$	$\frac{B_{mm}\cdot 10^{-3}{}^b}{\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}}$
B_{00}	-0.270922	0.53203
B_{10}	0.198028	-1.2710
B_{11}	0.237997	-1.3923
B_{20}		0.85773
B_{21}		3.8289
B_{22}		0.87685
B_{30}		
B_{31}		-2.2843
B_{32}		-1.9658
σ	30	0.16

^a Values for H_m^E . ^b Values for $C_{p,m}^E$.

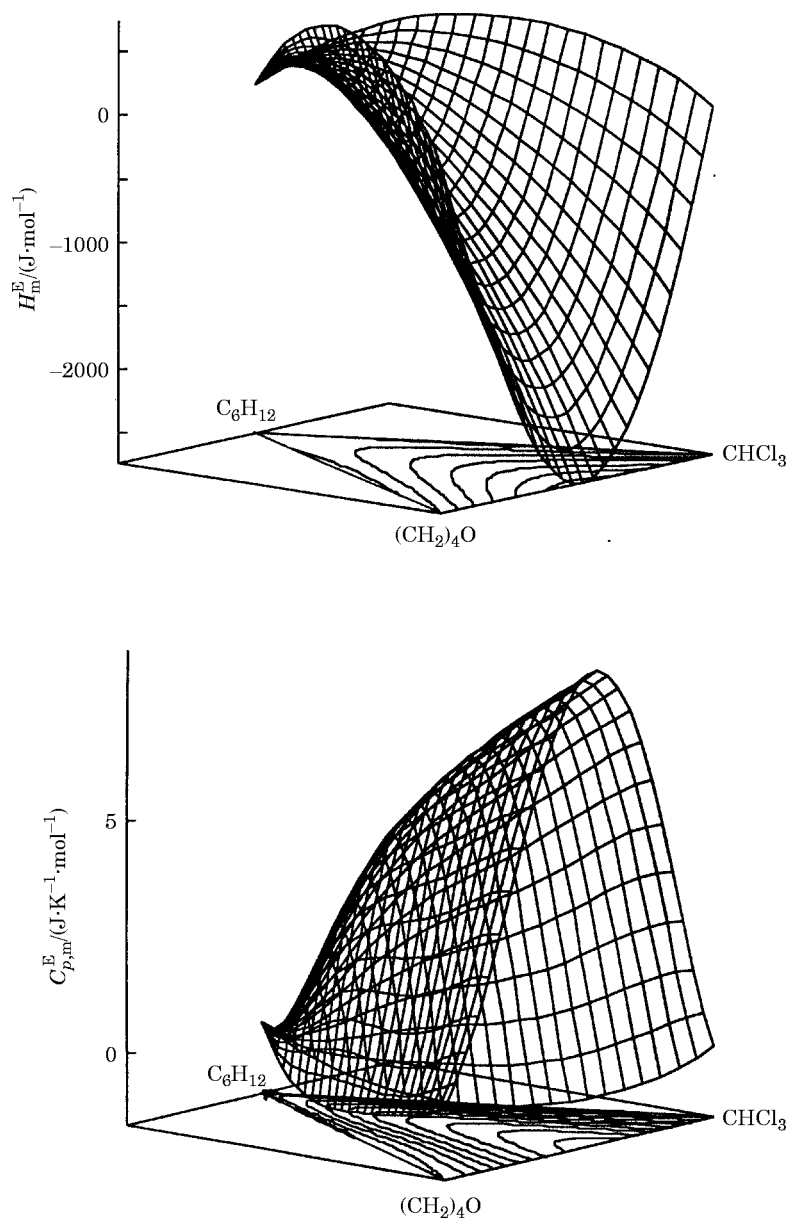


FIGURE 3. Excess molar enthalpies H_m^E and excess molar heat capacities $C_{p,m}^E$ for $\{x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_4\text{O} + (1 - x_1 - x_2)c\text{-C}_6\text{H}_{12}\}$ at $T = 298.15\text{ K}$: best-fitting surfaces from equation 2.

and 1–3,⁽¹⁸⁾ thus stressing the role of component designated as number 1, usually the more polar one.

The predictive ability of these methods is shown for both thermodynamic properties in figure 4. For asymmetric equations the minimum deviation is reported

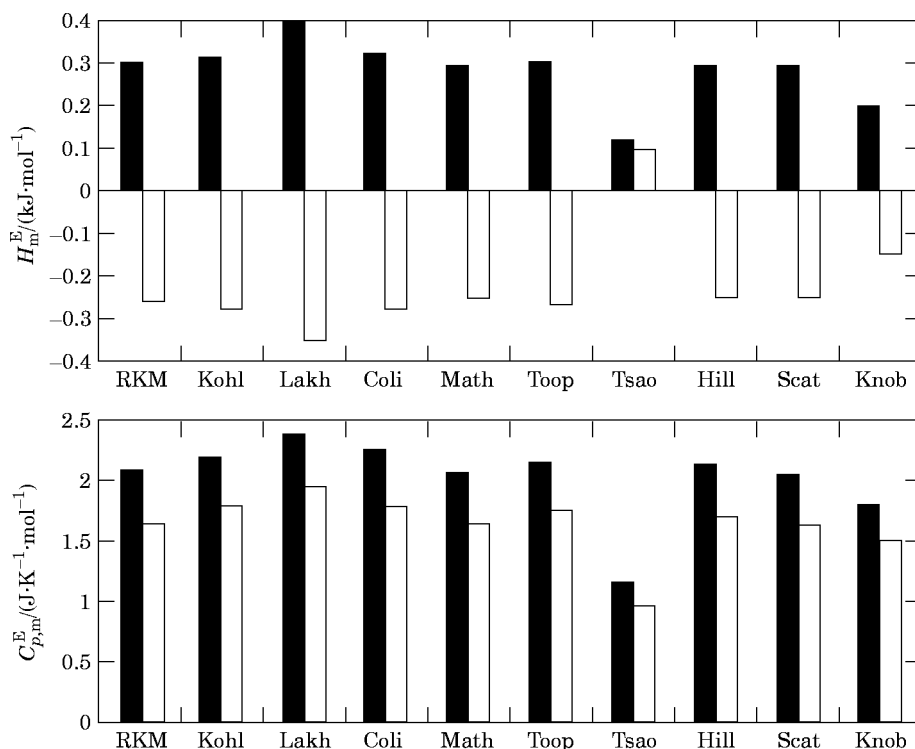


FIGURE 4. Comparison between the experimental data of H_m^E and $C_{p,m}^E$ and values calculated using empirical rules for $\{x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_6\text{O} + (1 - x_1 - x_2)c\text{-C}_6\text{H}_{12}\}$. ■, r.m.s.d. root mean square deviation; □, m.d. mean deviation (experimental—calculated). Methods from the left side: RKM = Redlich–Kister–Muggianu; Kohl = Kohler; Lakh = Lakhanpal; Coli = Colinet; Math = Mathieson and Tynne; Toop = Toop; Tsao = Tsao and Smith; Hill = Hillert; Scat = Scatchard; Knob = Knobloch and Schwartz. For the description of the methods see references 17 and 18.

by considering $c\text{-C}_6\text{H}_{12}$ as component 1 in all cases. The practical equivalence of the magnitude of the r.m.s. deviation and the mean deviation, both about ten times larger than the standard deviation of the fit represented by equation (2) (see table 6), indicates systematic displacements of the calculated values from the experimental data, the latter being larger in magnitude than the predicted $C_{p,m}^E$ but lower than the predicted H_m^E .

The r.m.s. deviation values did not generally change much with numbering of the components, indicating that asymmetric equations yield predictions comparable with symmetric ones. The only asymmetric method which displays a clear sensitivity towards a different designation of the components is that of Tsao and Smith. This method yields sensibly lower r.m.s. deviation values for H_m^E and $C_{p,m}^E$, by comparison with both symmetric and asymmetric methods, when component 1 is the non-polar C_6H_{12} . This is probably due to the different weight that this method ascribes, compared with other asymmetric methods, to

TABLE 7. Parameters of the UNIQUAC and Wang models for H_m^E

Mixture	$i-j$	UNIQUAC ^a			WANG ^b		
		δ_{ij}	δ_{ij}^*	σ ^c	α_{ii}	Δe_{ij}	σ ^c
$x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}$	1-2	0.64860	-4.5062	44	0.24748	8815.23	25
	2-1	-2.0000	0.41353			-5815.69	
$x\text{CHCl}_3 + (1-x)c\text{-C}_6\text{H}_{12}$	1-3	1.41510	0.22543	4.1	4.78937	439.656	4.5
	3-1	0.60767	0.55351			349.442	
$xc\text{-(CH}_2)_4\text{O} + (1-x)c\text{-C}_6\text{H}_{12}$	2-3	-0.29560	0.46800	2.0	2.93349	484.389	3.5
	3-2	1.8741	0.14420			267.991	
$x_1\text{CHCl}_3 + x_2c\text{-(CH}_2)_4\text{O} + (1-x_1-x_2)c\text{-C}_6\text{H}_{12}$				43 ^d			82 ^d

^a Molecular parameters q_i of UNIQUAC equation were 1.91, 1.76 and 2.38 for CHCl_3 , $c\text{-(CH}_2)_4\text{O}$ and $c\text{-C}_6\text{H}_{12}$, respectively; δ interaction parameters refer to the UNIQUAC expression as in reference 1.

^b Molecular collision diameters σ_{ij} of Wang equation were taken as being proportional to the cubic root of volume UNIQUAC parameters r_i , taking r_i of the above compounds, in the order, as 2.27, 2.35 and 3.15.

^c Standard deviation of the fit ($\text{J}\cdot\text{mol}^{-1}$).

^d Root mean square deviation.

$\{x\text{CHCl}_3 + (1-x)c\text{-C}_4\text{H}_8\text{O}\}$, *i.e.*, the mixture characterized by the larger (in magnitude) enthalpy and heat capacity values.

In any case, the predictive ability of the empirical procedures considered here for the ternary mixtures seems generally unsatisfactory. An extensive test of the same empirical equations on the excess enthalpies of a large set of ternary systems, containing compounds of very different nature,⁽¹⁸⁾ yields average r.m.s. deviations smaller than those observed here. However, if we make this comparison with the only system (chloroform + pyridine + acetic acid) characterized by analogous hydrogen bonds and analogous magnitude of the H_m^E values, the deviations are found to be even larger (r.m.s. deviation $\approx 460 \text{ J}\cdot\text{mol}^{-1}$ for symmetric and $\approx 260 \text{ J}\cdot\text{mol}^{-1}$ for asymmetric equations) confirming the inadequacy of empirical predictive rules when applied to such strongly asymmetric systems.

As far as the semi-empirical models are concerned, we have calculated the excess enthalpies of the ternary mixtures using the UNIQUAC,⁽¹⁹⁾ Wang,⁽²⁰⁾ and the UNIFAC⁽²¹⁾ group contribution method. The values of the parameters employed for the description of binary mixtures with the UNIQUAC and Wang methods are reported in table 7. Figure 5 compares the experimental data with values predicted by these models for the pseudobinary mixtures which are obtained by adding $c\text{-C}_6\text{H}_{12}$ to $(0.5 \text{ CHCl}_3 + 0.5 c\text{-C}_4\text{H}_8\text{O})$. The experimental data are represented by smoothed values calculated from equation (2), using the parameters of tables (3) and (6). The values calculated with the Muggianu's equation are also shown, since this method is commonly used by most authors and is representative of the average predictive ability of empirical methods.

Figure 5 shows that the theoretical models provide a better approximation for the experimental mixing enthalpies. The predicted values differ from the experimental ones by $\approx 200 \text{ J}\cdot\text{mol}^{-1}$ for the UNIFAC method; $\approx 100 \text{ J}\cdot\text{mol}^{-1}$ for the Wang method; and are almost coincident with them for the UNIQUAC method, despite the unsatisfactory description ($\sigma = 44 \text{ J}\cdot\text{mol}^{-1}$) of

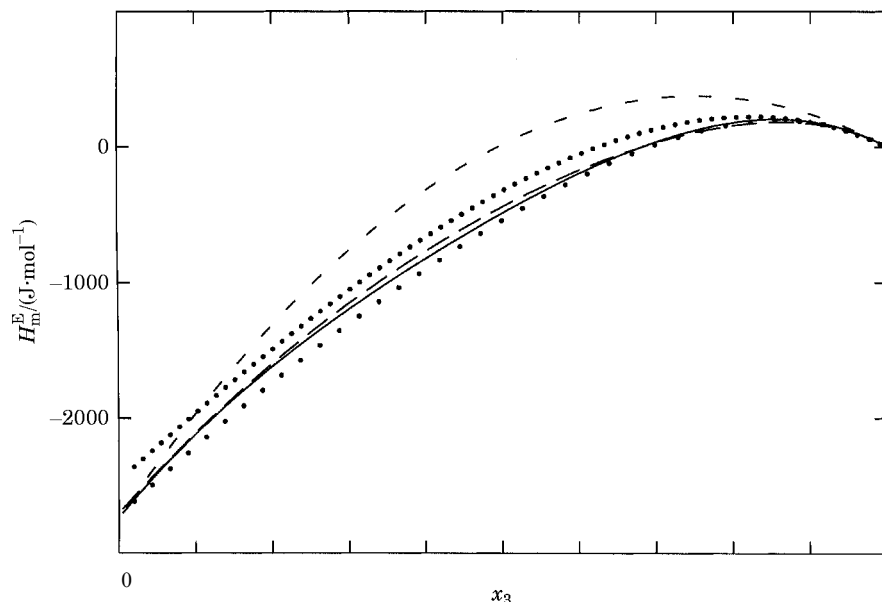


FIGURE 5. Comparison between the experimental H_m^E values and those predicted by various models. The curves refer to ternary mixtures obtained by adding c -C₆H₁₂ to (0.5 CHCl₃ + 0.5 c -(CH₂)₄O): —, experimental (equation 2); ●●●●, UNIFAC; ---, UNIQUAC; ○●●○, Wang; —×—, Muggianu.

$\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$ by the UNIQUAC method. The displacement of the UNIFAC function is larger at very low concentrations of the hydrocarbon, indicating an unsatisfactory prediction of the H_m^E values of $\{x\text{CHCl}_3 + (1-x)c\text{-(CH}_2)_4\text{O}\}$, which could be ascribed to an insufficient definition of the interaction parameters between CHCl₃ and the cyclic ether oxygen used in the UNIFAC database.

The data on the excess volumes and excess Gibbs energies will be reported in future papers.

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